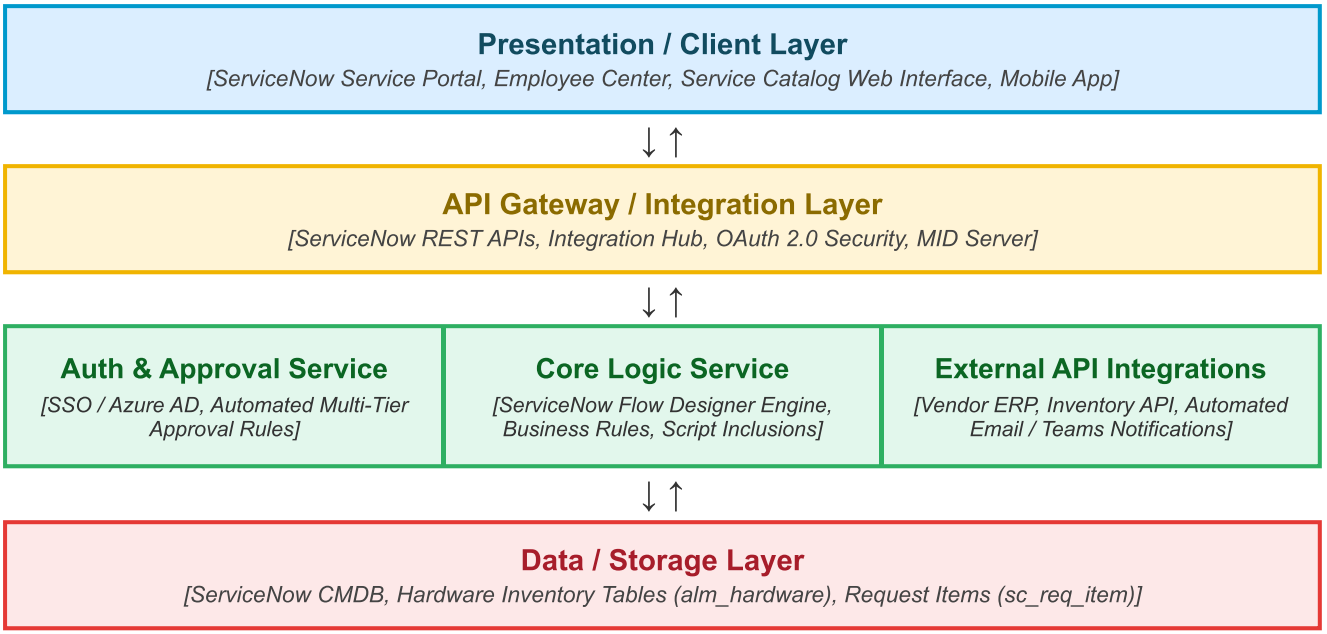
Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-6686
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Team Details	**Team Size:** 4 **Team Leader:** MANISHA R **Team Members:** • GOPIKA K • VINDHIYA G • NITHIYASRI M
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the self-service ordering interface for employees and managers to submit laptop orders, track status, and view pending approvals.	ServiceNow Employee Center, Service Portal, Service Catalog Widgets
API Gateway / Integration Layer	Secures and routes incoming/outgoing service requests, managing API authentication and connecting external ERP or vendor platforms.	ServiceNow REST API, Integration Hub, OAuth 2.0, MID Server
Auth Service	Handles user authentication, single sign-on (SSO), role-based access control (RBAC), and dynamic manager approval hierarchy lookup.	Azure Active Directory, Single Sign-On (SSO), ServiceNow User Roles
Core Logic Service	Executes the primary workflow logic, evaluating procurement rules, triggering approval tasks, and automating fulfillment catalog tasks.	ServiceNow Flow Designer, Business Rules, Flow Actions, JavaScript
External API Integrations	Sends real-time email/Teams notifications and connects with OEM procurement endpoints for automatic stock replenishment.	Microsoft Teams Spoke, Email Engine, REST Web Services
Data / Storage Layer	Stores information related to service requests, hardware asset inventory, approval histories, and user profile data.	ServiceNow CMDB, Asset Management DB (`alm_hardware`), MySQL Engine